EXPLORING FACTORS AFFECTING STROLLER SUITABILITY FOR CHILDREN WITH AUTISM USING MACHINE LEARNING-BASED QUESTIONNAIRE ANALYSIS

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ABSTRACT

This paper discusses how strollers suit children with Autism Spectrum Disorder (ASD) through analysis of a questionnaire based on machine learning. Data about child features, sensory sensitivities, caregiver stress, technology acceptance, and stroller use were gathered and evaluated by the caregivers. Gradient Boosting, Random Forest and Logistic Regression data preprocessing and machine learning methods were used to determine the most important predictors of stroller suitability, with class imbalanced using the SMOTE and further improved through hyperparameter tuning. Initially the accuracy for three models is similar, do not have a model outperform others. The findings reported that outdoor sensory reaction, age of manifestation of autism trait, the levels of stress of caregivers, and encounters with sensory calming intervention contributes greatly to the suitability of a stroller. After hyperparameter tuning, Gradient Boosting not only demonstrated the best predictive accuracy (86%) but also the most influential features and testified that the suitability of strollers is a multidimensional aspect. The most significant predictor of stroller suitability, according to feature importance and SHAP analyses, is outdoor sensory reactions (sensoryoutdoor). This is followed by age of autism trait onset (autismtraitage), caregiver stress-related variables (stresseffort, stressstressed, stresshausted, stressanxious, and stressavoid), sensory sensitivity (sensorylight), and experience with sensory-calming environments, including Snoezelen-related features. The results are practical in developing sensory-responsive strollers that will promote comfort, security, and emotional well-being among ASD children and assist the caregivers in their everyday outdoor endeavors.

Keywords: Autism Disorder, Sensory Sensitivity, Caregiver Stress, Stroller Suitability, Machine Learning, Gradient Boosting, Assistive Technology.

ABSTRAK

Kajian ini meneroka faktor-faktor yang mempengaruhi kesesuaian kereta sorong untuk kanak-kanak dengan Gangguan Spektrum Autisme (ASD) menggunakan analisis soal selidik berasaskan pembelajaran mesin. Data yang dilaporkan oleh penjaga mengenai ciri kanak-kanak, sensitiviti deria, tahap tekanan penjaga, penerimaan teknologi, dan penggunaan kereta sorong dikumpulkan dan dianalisis. Teknik penyediaan data dan pembelajaran mesin, termasuk Gradient Boosting, Random Forest, dan Logistic Regression, digunakan untuk mengenal pasti peramal utama kesesuaian kereta sorong, dengan ketidakseimbangan kelas menggunakan SMOTE dan dipertingkatkan lagi melalui penalaan hiperparameter. Hasil menunjukkan bahawa tindak balas deria di luar rumah, umur manifestasi ciri autisme, tahap tekanan penjaga, dan pengalaman dengan intervensi menenangkan deria memberi kesan ketara terhadap kesesuaian kereta sorong. Selepas penyetelan hiperparameter, Gradient Boosting mencapai ketepatan ramalan tertinggi (86%) dan mengenal pasti ciri paling berpengaruh, menekankan sifat multidimensi kesesuaian kereta sorong. Peramal paling ketara bagi kesesuaian kereta sorong, menurut kepentingan ciri dan analisis SHAP, adalah tindak balas deria luar (sensoryoutdoor). Ini diikuti oleh umur kemunculan ciri autisme (autismtraitage), pemboleh ubah berkaitan tekanan penjaga (stresseffort, stressstressed, stresshausted, stressanxious, dan stressavoid), kepekaan deria (sensorylight), dan pengalaman dengan persekitaran menenangkan deria, termasuk ciri berkaitan Snoezelen. Penemuan ini memberikan panduan berguna untuk mereka bentuk kereta sorong sensori yang meningkatkan keselesaan, keselamatan, dan kesejahteraan emosi kanak-kanak dengan ASD serta menyokong penjaga dalam aktiviti harian di luar rumah.

Kata Kunci: Gangguan Autisme, Kepekaan Deria, Tekanan Penjaga, Kesesuaian Kereta Dorong, Pembelajaran Mesin, Gradient Boosting, Teknologi Bantuan.

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LIST OF ABBREVIATIONS

ASD	-	Autism Spectrum Disorder
CART	-	Classification and Regression Tree
EDA	-	Exploratory Data Analysis
GB	-	Gradient Boosting
LR	-	Logistic Regression
ML	-	Machine Learning
RF	-	Random Forest
SMOTE	-	Synthetic Minority Over-sampling Techniques
SMRs	-	Sensory Management Recommendation System

CHAPTER 1

INTRODUCTION

1.1 General Overview

Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder, which occurs in about one out of every 127 people in the world (World Health Organization, 2025). Children with ASD also have unusual sensory processing, which is characterized by excessive sensitivity or lack of responsiveness to stimuli, including light, sound, touch, and movement (Patil and Kaple, 2023). The manner in which children interpret sensory data can actually influence their daily living as shown in their ability to move about, their interaction and their enjoyment of outdoor activities. As an example, they can easily be disturbed by such things as bright lights, loud sounds, or crowded places, not only does this disturb the child, but it also makes things more difficult to the caregivers.

Although outdoor activities are useful in the physical and emotional growth of children, perceptual delays in children with ASD can be encountered as a result of the unpredictability of sensory inputs like noise, crowds, weather conditions and overloading of the sight. Such difficulties do not only influence the comfort and emotional control of the child, but also heighten physical and emotional overload on the caregivers. Thus, caregivers frequently face difficulties in searching the mobility solutions that would be safe and appropriate to meet the individual sensory and behavioral requirements of children with ASD.

Over the past few years, the use of questionnaires in data collection has been praised as one of the effective methods of gathering data about the caregiver observations concerning their sensorial sensitivities, behavioral reactions and mobility

experiences among children. Such information, coupled with analytical methods that are based on empirical data, allows exploring factors that determine the suitability of such assistive tools in a systematic manner. The methods of machine learning, specifically, offer the possibility to evaluate more complicated relationships between several variables and reveal patterns that might be not easy to identify using the traditional analysis.

The idea of the sensory smart stroller is one of the new assistive solutions. In contrast to traditional strollers, sensory smart strollers can be designed with alternatives that include adjustable lighting, soothing sound, vibration control, environmental monitors, and caregiver-controlled parameters to make them more comfortable and safe in the outdoor environment (Walker, 2025). Nevertheless, the efficiency of these strollers is heavily dependent on the ability of their design to correspond to the real needs and experiences of children with ASD and their caregivers.

The research in this work is based on the analysis of the caregiver-reported questionnaire data and machine learning methods to investigate the most significant factors that determine the suitability of the strollers in children with autism. The findings will be used to design solutions to enhance outdoor experiences by children and caregivers by focusing on the most influential sensory, emotional and contextual variables to design data-driven and more informed strollers.

1.2 Problem Background / Research Background

The children with ASD normally respond to the sensory experiences in a manner that is very different than what we as human beings would have anticipated. There are those that are quite sensitive to bright light, loud sounds, touch or even movement. Some other people may react very little to some stimuli (Centers for Disease Control and Prevention, 2025). Such variations may result in them finding it difficult to participate in the daily life and may also complicate their locomotion. Normal strollers, those that are targeted to children with no recorded developmental problems normally do not accommodate these sensibilities.

Traditional strollers are normally created to suit the needs of children who are developmentally and sensually normal and these differences are not reflected. Subsequently, the children with ASD might feel uneasy, anxious, or overstimulated by a stroller, which causes behavioral distress and the decreased readiness to participate in outdoor activities. It, in its turn, increases the pressure on the caregivers, who have to deal with both the environmental needs and the sensory requirements of the child at the same time (Johansen et al., 2016).

Even though assistive mobility technologies have been studied in older children and adults (wheelchair and adaptive devices), the design of strollers among children with autism of younger age has not received enough research interest. The available solutions of smart mobility are usually focused on mechanical performance or convenience, but not sensory adaptability and caregiver-child interaction. Therefore, the prospect of the suitability of strollers among children with ASD is not fully covered.

Most recent developments in machine learning have shown potential in interpreting caregiver-reported data to detect behavioral and sensory patterns (Madububambachu et al., 2024; Alateyat et al., 2022). However, a significant part of the current literature is devoted to the diagnosis, or behavioral testing, or adult assistive technologies. The research gap is evident in the fact that there is no direct correlation between the answers of the caregivers in the questionnaires and the analysis of the stroller suitability and design parameters.

To fill this gap, it is important to take a systematic method that will incorporate both questionnaire-based data gathering and machine learning tools in order to investigate the combined effects of sensory sensitivities, emotional regulation, stressor of the caregivers, exposure to the environment, and perception of technology on the suitability of strollers. It is necessary to comprehend these relationships in order to come up with stroller solutions that are not only workable but also sensitive to the lived experiences of children with ASD and their caregivers.

1.3 Research Objectives

This paper is aimed at investigating and finding the key aspects that determine stroller suitability among children with Autism Spectrum Disorder (ASD) based on the data provided in the caregiver-reported questionnaire and machine learning. In particular, the study will focus on the creation of a structured and standardized questionnaire to reflect the observations of caregivers in terms of sensory sensitivities in children, their emotional regulation, mobility issues as well as outdoor life. The data gathered are washed, treated and ready to be analyzed through machine learning to make sure that the results are consistent and reliable. Logistic Regression, Random Forest, and Gradient Boosting are the several machine learning models applied and compared to investigate their strength in exploring factors influencing stroller suitability. Secondly, the research also aims to evaluate the importance of features in order to establish which sensory, behavioral, and contextual variables have the strongest impact on the stroller suitability results, and therefore can give data-driven conclusions to guide the construction of sensory-responsive smart strollers.

1.4 Scope

This paper will discuss the children aged under thirteen years with Autism Spectrum Disorder and factors applied to their appropriateness to use in a stroller during outdoor activities. The margin is restricted to the questionnaire respondents that are caregivers and will include data on child demographics, sensory sensitivities, emotional co-regulation, caregiver stress, stroller use patterns, and the perception of assistive stroller capabilities. The use of machine learning methods is only used to examine the association and relationships between these factors and stroller suitability. In the study, no clinical diagnoses, treatment interventions, or physical modifications to the available stroller products are sought. Instead, it puts an emphasis on the realization of behavioral and sensory-related variables that will determine the stroller suitability, and the aim was to provide evidence-based design-related variables in the

future development of smart strollers and enhancement of caregiver and child outdoor experience.

1.5 Importance/Significance of Project

It is also important to learn about the determinants of stroller appropriateness to children with ASD to improve their comfort, safety, and engagement in outdoor activities. The study is important as it will fill the gap between the experiences of caregivers and the objective data analysis, allowing to find out the sensory, emotional, and contextual aspects of the problem of stroller appropriateness. The research offers objective data through the application of machine learning models which can be utilized to design assistive technologies in a personalized manner instead of designing solutions that are applicable to everyone. The results have practical implications to caregivers to alleviate stress during outings, the designers with sensory-conscious features of a stroller, and researchers with proof of the usefulness of questionnaire-based machine learning analysis in assistive technology studies. Furthermore, the research will be beneficial to the general society by creating awareness of the need to have more inclusive mobility solutions that will support children with autism and their families.

1.6 Report Organization

Table 1.1 Report Organization

Chapter	Content
1	Provides background on autism spectrum disorder and sensory challenges, states the problem, research questions and objectives, scope, significance, and report organization.
2	Reviews previous studies on sensory processing in children with autism, questionnaire-based data collection, and the application of machine learning for exploring stroller suitability factors.
3	Describes the research design, participant selection, questionnaire structure, data collection procedures, data preprocessing, and machine learning model implementation.
4	Presents descriptive analysis of the questionnaire dataset, including demographic summaries, sensory profiles, stroller usage patterns, and preliminary visualization of key variables to identify trends, distributions, and potential correlations.
5	Discusses the application of machine learning models, evaluates model performance using appropriate metrics, interprets feature importance, and identifies key factors for stroller suitability.
6	Summarizes the study, highlights implications for stroller design, provides recommendations for caregivers, designers, and researchers, and outlines limitations and directions for future research.

1.7 Conclusion

Autistic children also have serious sensory and emotional issues that influence their mobility and outdoor activities. Traditional strollers do not usually meet these requirements, and children experience discomfort, whereas caregivers become more stressed. The paper will use a questionnaire-based, machine learning method to examine the variables that define the suitability of a stroller to ASD children. The research will help address the creation of more flexible and caregiver-informed stroller designs that would help to improve the comfort, safety, and well-being of autistic families, identifying major factors that contribute to sensory, emotional, and contextual aspects.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Autism Spectrum Disorder (ASD) is a complicated neurodevelopment disorder, which is marked by difficulties in social interaction, communication, and behavioural difficulties that are usually complemented by unusual sensory processing. Children with ASD are often hyper sensitive or under responsive to various stimuli (sound, light, touch, and motion) and this can have a profound effect on their everyday functioning and mobility (Centers for Disease Control and Prevention, 2025). Sensory processing may lead to distress in normal activities, such as outdoor outings and create further problems to the caregiver in terms of assuring the comfort and safety of the child.

Recent developments on machine learning have demonstrated significant possibilities on exploring suitability factors relying on questionnaire-based and observational data. The models may be used to offer information about the sensory profile of a child to design assistive technologies that would be personalized to the needs of the person (Madububambachu, Ukpebor, and Ihezue, 2024). It has proven that multiple machine learning algorithms, such as Random Forest, Gradient Boosting, or Support Vector Machines, can make exploration stroller suitability factors (Alateyat et al., 2022).

This literature review is aimed at investigating the literature on sensory processing in children with ASD, the existing condition of assistive mobility devices, and the use of machine learning techniques in exploring suitability factors. By so

doing, it establishes knowledge gaps in literature that support the necessity of formulating questionnaire-based machine learning models to inform the design of sensory smart strollers to children with autism.

2.2 Sensory Processing in Children with Autism

Sensory processing, in which individuals receive, interpret and respond to stimuli from their environment, is profoundly different for many children with Autism Spectrum Disorder (ASD) compared with their typically-developing peers. Understanding these differences is essential when designing mobility aids, such as strollers, that must function in real-world, multisensory environments.

2.2.1 Prevalence and Patterns of Sensory Differences

Research shows that atypical sensory responses are highly prevalent in children with ASD. For example, a comparative study found that children with ASD exhibited significantly higher rates of hyper-responsiveness, hypo-responsiveness and sensory-seeking behaviours across auditory, tactile, visual and proprioceptive domains, when compared to non-ASD children ($p < 0.001$ for all domains) (Maqsood, Afzal, Aftab, & Ahmad, 2025). Another large-scale review highlighted that sensory hyporesponsiveness was strongly correlated with core features of ASD, and sensory hyperresponsiveness was associated with social awareness difficulties and affective disorders (Glod, Riby, Honey, & Rodgers, 2015).

A foundational neurophysiologic review noted that atypical sensory-based behaviours are “ubiquitous” in ASD, and that children and adolescents with ASD show distinct neural responses to visual, tactile and auditory stimuli compared with controls (Marco, Hinkley, Hill, et al., 2011).

2.2.2 Impacts on Daily Functioning and Mobility

These differences in sensory processing do not happen in a vacuum - they have significant effects on the functioning and the ability of children to be involved in daily mobility and outdoor activities. An example of this is hyper responsiveness towards lights or abrupt noises that may result in emotional breakdown or anxious behaviour as one moves out of a home environment into a busy and loud outdoor setting. The hypo responsiveness may result in delayed responsiveness or require excessive stimulation such as movement, vibration, not possible by conventional mobility devices.

Having said that, a longitudinal study discovered that initial high scores on hypo responsiveness and sensory seeking were related negatively to subsequent adaptive behaviour in vulnerable children, indicating that unresolved issues with sensory processing may accumulate into functional impairments (Gunderson et al., 2024).

2.2.3 Relevance to Assistive Mobility and Design

To designers of mobility imply various important considerations, these patterns of sensory processing imply. When a child happens to be easily overstimulated by the surrounding light or sound, then a regular stroller canopy or structure may not protect or control such stimuli. When a child requires movement or pressure (sensory seeking behaviour), mobility aid that is excessively rigid or one that does not feature dynamically might not allow the comfort of the child. The literature highlights the fact that mainstream equipment hardly ever takes into consideration the broad range of variation of sensory profiles of children with ASD, but these profiles do not lack real behavioural and physiological implications.

2.3 Assistive Mobility and Smart Stroller Technologies

Over the past several years, research papers have shown an increasing interest in the development of infant care products that include smart technologies to improve the safety and comfort of children, as well as minimize the activity of caregivers. Such innovations are automatic cradles, responsive bassinets and sensor-based strollers which are attuned to the environment and needs of the child.

2.3.1 Automatic and Responsive Baby Cradles

A number of studies have shown the possibility of using motorized systems in order to offer comfort and responsiveness to babies. Kadu et al. (2021) suggested that a DC motor provides rotational movement to a bassinet based on a microcontroller-programmed protocol. The motor will spin clockwise over a predetermined time to propel the bassinet forward and counterclockwise to propel it sideways in an endless, precision movement that comforts the infant (Kadu et al., 2021).

On the same note, Goyal et al. (2020) came up with an E-Baby Cradle that swings automatically when a baby cries. The system has a cry-analyzing module, which identifies the voice of the infant and initiates the cradle action until the baby falls to sleep. Cradle speed can be altered depending on the preference of the user. The system also includes an inbuilt alarm which notifies two situations: first, when the mattress is wet to ensure hygiene and second, when the infant fails to cease crying after a certain amount of time, and this notifies the caregivers to attend to the child (Goyal et al., 2020).

Chao et al. (2019) introduced the further refined automated cradles such as resonant electric cradle design, which has sensors of oscillation and cry detection. This system applies the force at the critical point of oscillation and the motion is most efficient and consumes less energy. The resonance principle enables the cradle to have the best soothing effects using less power (Chao et al., 2019).

2.3.2 Sensor-Based Smart Strollers

Automated mobility has now been applied to strollers also. Almutairi et al. (2022) introduced an automatic stroller with sensors to identify the obstacles and prevent accidents, like holes in the outdoor use, thus improving safety. The stroller is also connected to a mobile application that can track its position and inform the caregivers about it in real-time, which guarantees the safety and convenience (Almutairi et al., 2022). Furthermore, Davis et al. (2021) have created a smart braking system to strollers or rather utilizes capacitive touch sensors, a relation mechanism, and a 12 V solenoid to provide quick and trustworthy stopping once needed. They show that it is possible to combine various sensing and actuation technologies into strollers to enhance user experience and safety (Davis et al., 2021).

2.3.3 Implications for Sensory Smart Stroller Design

All these studies point to the fact that the incorporation of sensor-based, automation, and feedback into the infant care products can greatly enhance comfort, safety, and efficiency of the caregivers. Nevertheless, the majority of studies have concentrated on the use of general infant strollers or cradles as opposed to products that are specifically designed to suit autistic children. Considering sensory sensitivities of ASD, there is still a gap in designing strollers which should be not only physically safe but also emotionally and sensually responsive. Using questionnaire-based models, sensor-based actuation can be a viable possibility in developing personalized, sensory-responsive smart strollers to children on the spectrum.

2.4 Machine Learning Applications in Sensory and Mobility Assistive Technologies for ASD

Machine learning (ML) methods played a central role in the last few years, as machines improve the creation of assistive technologies that could be more helpful to people with neurodevelopmental disorders, including Autism Spectrum Disorder (ASD). They are used in areas of early diagnosis up to taking a step ahead to anticipate

sensory requirements and adaptability mobility aids, however the latter is less investigated.

2.4.1 Machine Learning in Assistive Mobility Device Assessment

ML though unrelated with the stroller design, has been used to evaluate and prescribe relevant mobility devices to other groups of people. Indicatively, one study applied classification and regression tree (CART) models and a naive bayes classifier to diagnostic records of wheelchair users and found essential factors such as head control, pelvic condition, cognitive and judgment ability and age. Accuracy of the model was 72 in identification of the type of devices (Martins, Santos, Frizera-Neto, and Ceres, 2012).

Similarly, a review on assistive mobility devices noted that smart walkers and other adaptive devices increasingly incorporate sensors and algorithms to assist users with mobility limitations (Fang & Ping, 2022). These findings suggest that ML can support device-recommendation systems and real-time adaptive functions, setting a precedent for stroller innovation.

2.4.2 Machine Learning in Sensory Management Systems for Children with ASD

More directly relevant to this study are works that use ML combined with sensory profiling and real-time environmental data to support children with ASD. Deng and Rattadilok (2022) developed a “Sensory Management Recommendation System” (SMRS) that integrated caregiver questionnaire responses, wearable sensors like heart rate, skin conductivity, motion, and environmental sensors like noise, brightness, temperature. The system applied ML to detect distractions or anxious states and used fuzzy logic to recommend strategies to calm the child. The authors report positive impacts on attention and stress levels in classroom settings (Deng & Rattadilok, 2022).

The success of such systems demonstrates that ML models can meaningfully process heterogeneous data (questionnaire + sensor) to predict states relevant to sensory management. Translating this to stroller design implies that ML could predict a child's sensory state or trigger for event like overstimulation and then adapt stroller features (lighting, sound, vibration) accordingly.

2.4.3 Gaps in Stroller-specific Applications

Despite these advances, there is a notable gap: very few studies apply ML to design or adapt mobility devices specifically for children with ASD, especially strollers that accommodate sensory dynamics rather than only physical mobility. Existing works focus more on diagnosis, behaviour recognition, or assistive devices for other groups. For example, while the SMRS discussed above is helpful, it is classroom-oriented rather than outdoors/stroller-oriented. Meanwhile ML models in mobility aids often focus on adults or older children with physical disabilities rather than sensory-sensitive younger children.

2.4.4 Opportunities for Research

The current research has a number of opportunities with this research gap. Firstly, machine learning techniques may be applied to the data of the caregiver questionnaire to determine the main factors that affect the suitability of a stroller with children having autism when they are outdoors, in terms of sensual sensitivities and experiences of outdoor mobility. The research has the potential to forecast the stroller related needs by using information provided by the caregiver on the observations of the needs of the correct lighting conditions, the need to have appropriate lighting settings and needs to have sound control and the need to have movement stability. Second, the use of questionnaires as the basis of prediction can be conceptually connected to the ability of smart strollers, whereby the results of the survey can guide adaptive stroller. This forms a basis towards the systems of the future where the stroller responses are matched with the anticipated child requirement. Lastly, it is crucial to make machine learning models interpretable, and the caregivers and designers should have a clear picture of how the identified sensory sensitivities are associated with the

stroller suitability features. This interpretability gives a boost to trust and facilitates the application in stroller design decisions.

2.4.5 Summary

Assistive technologies using machine learning on children with autism have proven to be highly effective in eliciting any meaningful pattern in the questionnaire based data. According to previous research, sensory and behavioral data, which are reported by the caregivers, can be successfully processed to assist in the development of individual mobility solutions. Nonetheless, very little has been done to apply such machine learning-propelled questionnaire analysis to analyze the suitability of strollers to children with autism. This shows the topicality and novelty of the given research, as its subject is to find out and evaluate factors that affect stroller suitability with the help of machine learning algorithms. The research design based on feedback of caretakers helps to advance data informed stroller design indicative of the address of sensory comfort, safety and usability of children to autism.

2.5 Conclusion

The chapter has had a literature review associated with what has been known on sensory processing among children with Autism Spectrum Disorder, assistive mobility technologies such as stroller systems, and the use of machine learning in comprehending sensory and behavioural patterns. Previous researches always indicate that unusual sense responses are typical of children with ASD and impact a lot on their behaviour, human activities, and comfort in outdoor activities. The properties of sensory attributes have the impact on the way children perceive and interact with the surrounding and the significance of the sensory factors in assessing the suitability of the stasters should not be overlooked.

The analysis of the assistive mobility technology shows that there has been significant advancement in the creation of smart strollers and sensor-driven mobility aids especially in aspects to do with safety, monitoring, and movement management. Nevertheless, the majority of solutions that have been developed are still general in their nature and are not developed specifically to cater to the sensory sensitivities that come with autism. Aspects associated with sensory comfort, emotional control, and customized environmental adaptation are not well represented, even though they are significant when assisting children with ASD to engage in mobility activities.

The sources covering the topic of machine learning show that it has a great potential in processing and analysing questionnaire data in order to forecast sensory requirements and behavioural trends. Research has demonstrated that self-reported information provided by caregivers is useful and can be utilized to simulate individual sensory profiles and aid in individual intervention. In spite of these progresses, an apparent gap in the research is the implementation of the questionnaire analysis using machine learning concerning the evaluation of the suitability of strollers in children with ASD. Existing studies seldom connect the perspectives of caregivers with design concerns of a stroller or even analyse the role of sensory elements in the suitability of stroller features.

In general, the literature confirms the need and the possibility of researching the suitability of strollers using the data-driven approach with a focus on the sensory needs. The gap in the present study is bridged by examining the caregiver-reported factors with machine learning methods to gain a better insight on the role of sensory properties on the suitability of a stroller to children with autism. These results of this chapter are the theoretical and empirical basis of the research methodology in the following chapter.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Research Design

The research design of this paper provides the use of a quantitative research design to gather survey-based data (caregiver-reported) on the issues related to children with Autism Spectrum Disorder ASD and their experiences with the use of strollers. The study aims at finding out how sensory sensitivities, emotional regulation, stress caused by caregivers and environmental influences affect the stroller suitability.

The paper also uses machine learning to examine the data gathered and identify patterns and relationships which would otherwise be difficult to identify using conventional statistical analysis. This will enable objective determination of factors that have a major influence on stroller suitability, which can be used to inform the creation of sensory-responsive strollers.

Particularly, the research design involves:

- Questionnaire design to include observations by the caregivers of the sensory and behavioral responses of their child.

- Caregiver-collected data of children with ASD who were below the age of thirteen years.

- Preprocessing Data cleaning and preparing the dataset to be analyzed using machine learning.

- The use of several machine learning models, such as Logistic Regression, Random Forest, and Gradient Boosting, to determine the most significant factors that affect the stroller suitability.

- Analysis of feature importance to rank and explain the effects of every factor on outcomes of stroller suitability.

This design will guarantee a systematic and rigorous approach to comprehending stroller suitability coupled with interpreting caregiver experiences with machine learning analysis to get evidence based insights.

3.2 Data Collection & Participant Selection

The study targets caregivers of children diagnosed with Autism Spectrum Disorder (ASD) under the age of thirteen. The goal is to gather reliable and relevant data on the child's sensory sensitivities, behavioral responses, stroller usage, and caregiver perceptions of assistive features.

3.2.1 Participant Selection

Inclusion Criteria:

- Child is diagnosed with ASD formally.

- Child has an age of less than thirteen years.

- Caregiver will oversee the child whenever outdoor activities are taking place.

- Caregiver will be willing to fill in the questionnaire willingly and give the correct responses.

Exclusion Criteria:

- Children between the ages of thirteen and more.
- Nurses who cannot give full or dependable answers.
- Children who have other medical conditions that contribute greatly to immobility other than ASD.

The target population of 300 caregivers was established to provide adequate information to analyze with machine learning and at the same time makes it feasible. The sample was designed to represent a wide range of sensory sensitivities, behavioral tendencies and experience in the use of the strollers.

3.2.2 Questionnaire Design

The structured questionnaire was developed to capture caregiver-reported data across multiple domains relevant to stroller suitability for children with Autism Spectrum Disorder (ASD). The questionnaire consists of 13 parts, each focusing on a specific aspect of the child's behavior, sensory experiences, caregiver perceptions, or assistive technology features. Each part contains multiple sub-questions to obtain detailed insights.

Part 1: Child's Information

- Collects basic demographic information of the child, including age, gender, level of support needed, and age at which autistic traits were observed.

Part 2: Snoezelen Rooms (Calm Rooms)

- Measures caregiver perception of the child's use of Snoezelen rooms and their effectiveness.

Part 3: Stroller Use & Outdoor Activities

- Captures information on stroller usage patterns and outdoor activity experiences.

Part 4: Sensory Sensitivity

- Evaluates the child's responses to sensory stimuli such as light, sound, touch, and motion.

Part 5: Caregiver Stress

- Assesses the stress level of caregivers during outings with the child.

Part 6: Emotional Co-regulation

- Measures how caregivers and children manage emotions during outdoor activities.

Part 7: Self-Efficacy

- Captures caregiver confidence in managing child's sensory and behavioral challenges during outings.

Part 8: Cultural Acceptance and Social Identity

- Explores the influence of cultural norms and social identity on stroller use and caregiver behavior.

Part 9: Sensory Intervention, App Control, and Sensor Function

- Evaluates features of sensory smart strollers and their potential impact.

Part 10: Perceived Tool Utility

- Measures caregiver perception of the usefulness and practicality of the stroller's features.

Part 11: Sense of Control

- Captures caregivers' perception of control over child's sensory experiences during stroller use.

Part 12: Smart Trust Mechanism

- Assesses the degree of trust caregivers place in the stroller's automated or smart features.

Part 13: Willingness to Use and Pay & Caregiver Information

- Measures the caregiver's readiness to adopt sensory smart strollers and pay for enhanced features.

Each part of the questionnaire is designed to provide a comprehensive understanding of both the child's needs and the caregiver's perceptions, ensuring that collected data can be effectively analyzed using machine learning techniques to determine factors affecting stroller suitability.

3.3 Data Preparation and Preprocessing

The dataset collected from the autism survey comprised 300 responses with 84 variables, covering child demographics, caregiver information, sensory sensitivities, and perceptions of assistive technologies such as Snoezelen rooms and sensory smart strollers. Prior to analysis, the data underwent extensive cleaning, transformation, and encoding to ensure suitability for machine learning models.

3.3.1 Data Cleaning and Standardization

1. **Column Renaming:**
   All survey columns were renamed using consistent and meaningful variable names for clarity and ease of processing. For example, “Child gender” was renamed child_gender, “How often do you use a stroller when you take your child out?” became stroller_freq, and “I am willing to use a sensory smart stroller which can improve my child's comfort” was renamed willing_use.

2. **Whitespace and Typo Removal:**
   String variables were stripped of leading/trailing whitespace, and minor inconsistencies in categorical responses were corrected. Examples include:
   - Standardizing responses like 3-6 / Month' to 3-6 / Month
   - Standardizing income brackets and duration formats.

3. **Handling Missing Values:**
   Columns with missing or “Not sure” responses were carefully addressed. For the target variable stroller_suitable, responses marked as “Not sure” were removed to create a clear binary classification target.

4. **Duplicate Rows:**
   Duplicate entries were checked and removed where necessary to avoid bias in model training.

3.3.2 Feature Categorization

The survey questions were grouped into meaningful categories to guide feature selection:

1. **Child Information:** Age, gender, level of support, autism trait onset age, communication ability, mobility.
2. **Snoezelen Room Usage:** Frequency, effectiveness, accessibility, and cost perceptions.
3. **Stroller Use & Outdoor Activities:** Frequency, calming effect, suitability, outing duration, and child calm duration.
4. **Sensory Sensitivity:** Responses to light, noise, crowds, outdoor stimuli, and emotional meltdowns.
5. **Caregiver Stress:** Avoidance of outings, effort required, exhaustion, and anxiety levels.
6. **Emotional Co-Regulation & Self-Efficacy:** Ability to regulate own emotions, recognize child distress, and handle high-sensitivity situations.
7. **Social Perception & Cultural Acceptance:** Social judgment, pride, awareness, and community understanding.
8. **Stroller Features & App Functions:** Sensory, remote-control, smart sensors, emergency functions, and app features.
9. **Sensor Data:** Light, sound, temperature, motion, air quality, heart rate, and GPS.
10. **Ranking & Willingness:** Importance ranking of features, willingness to use, pay, and recommend.
11. **Caregiver Demographics:** Gender, age, education level, income, area, and relationship to the child.

3.3.3 Variable Encoding and Transformation

To prepare the data for machine learning:

1. **Target Variable:**
   stroller_suitable was binarized:
   - 1 = Yes (suitable)
   - 0 = No (not suitable)

2. **Categorical Variables:**
   One-hot or ordinal encoding was applied based on the variable type. For example:
   - child_gender: Male=1, Female=0
   - support_level: Not sure=0, L1=1, L2=2, L3=3
   - stroller_freq: Never=0, Occasionally=1, Often=2, Always=3

3. **Ordinal Variables:**
   Variables such as age brackets, calm duration, price ranges, and caregiver education were converted to numeric ordinal values reflecting progression or magnitude.

4. **Numerical Features:**
   Continuous or ordinal numeric features were scaled using StandardScaler, particularly for Logistic Regression.

3.3.4 Feature Selection

After encoding, open-ended features (open_features) were excluded from model training due to their textual nature. Correlation analysis was performed to identify features strongly associated with the target variable. This guided the selection of variables for inclusion in the final model.

3.3.5 Data Splitting and Balancing

- Training-Test Split:

The dataset was divided into 80% training and 20% testing, with stratification based on the target variable to maintain class balance.

- Handling Imbalanced Classes:

The training set was balanced using SMOTE (Synthetic Minority Over-sampling Technique)**,** ensuring both classes were adequately represented.

3.4 Machine Learning Model Implementation

The prepared dataset was used to develop predictive models for determining whether a stroller is suitable for a child with Autism Spectrum Disorder (ASD). Three supervised machine learning algorithms were selected based on their suitability for classification tasks and ability to handle nonlinear relationships: **L**ogistic Regression (LR), Random Forest (RF), and Gradient Boosting (GB)**.**

3.4.1 Model Selection

- **Logistic Regression:**

A baseline linear model used to estimate the probability of stroller suitability. It provides interpretability through feature coefficients. Standard scaling was applied to all features before training, due to the model's sensitivity to feature magnitude.

- **Random Forest:**

An ensemble tree-based method that constructs multiple decision trees and averages their predictions. It handles nonlinear relationships well and is robust to overfitting. Feature importance scores are easily interpretable.

- **Gradient Boosting:**

Another ensemble method that sequentially builds trees to correct errors from previous iterations. It is highly effective for imbalanced or complex datasets and provides detailed feature importance for model interpretability.

3.4.2 Model Training

- **Data Scaling:**

For Logistic Regression, all features were standardized using StandardScaler. Tree-based models (RF and GB) used the original feature values.

- **Class Balancing:**

The training set was oversampled using SMOTE (Synthetic Minority Over-sampling Technique) to mitigate class imbalance in the target variable stroller_suitable.

- **Model Fitting:**

Each model was trained on the oversampled training set. Hyperparameter tuning via Grid Search determined the optimal configurations for each algorithm.

3.4.3 Hyperparameter Tuning

To optimize model performance, Grid Search with 5-fold stratified cross-validation was conducted. The following hyperparameters were tuned:

- Logistic Regression: Regularization strength C and solver type (lbfgs, liblinear)
- Random Forest: Number of trees (n_estimators), maximum tree depth (max_depth), and minimum samples per split (min_samples_split)
- Gradient Boosting: Number of trees (n_estimators), learning rate (learning_rate), and maximum tree depth (max_depth)

The F1-weighted score was used as the evaluation metric during cross-validation to account for class imbalance.

3.4.4 Model Evaluation

Models were evaluated on the hold-out test set using the following metrics:

- **Accuracy:** Overall proportion of correct predictions
- **Precision, Recall, and F1-score:** To measure performance for both classes
- **Confusion Matrix:** To visualize true positives, false positives, true negatives, and false negatives

The evaluation provided insights into the ability of each model to classify stroller suitability correctly, and highlighted trade-offs between precision and recall.

3.4.5 Feature Importance Analysis

The scores of feature importance were calculated in order to determine what child attributes, sensory sensitivities, caregiver qualities, or stroller characteristics had the strongest impact on the target outcome. The impact of predictors on the model outcome was visualized through a heatmap of correlation and a bar chart of the 20 most important features to the model.

3.5 Conclusion

The data was used to build predictive models of stroller suitability that were dependable through meticulous preprocessing, balancing of the classes, hyper parameter optimization and model testing. These models will give us an insight of the most important factors that impact the decision making of the caregiver and they will identify important variables (child, sensory, and caregiver) that will be used to design and determine the functionality of the strollers.

CHAPTER 4

EDA & INITIAL RESULTS

4.1 Introduction

This chapter gives the findings of the data analysis and machine learning experiments in the course of this research. Through the analysis, the analysis will identify and assess the determining factors of suitability of the strollers to children with Autism Spectrum Disorders (ASD) using data on the questionnaire that will be reported by the caregivers. Statistical and machine learning methods were then used to analyze the associations between child characteristics, sensory sensitivities, caregiver stress, technology acceptance, and perceptions of stroller suitability using the processed survey responses.

To introduce the reader to the overall picture of the responses, the chapter starts with the descriptive and exploratory analysis to outline the dataset and bring out the general patterns in the responses. This is explained by correlation analysis to investigate the strength and direction of relations between the independent variables and stroller suitability. Then, the predictive performance of several machine learning models, namely, Logistic Regression, Random Forest, and Gradient Boosting, are assessed and compared. Lastly, the analysis of feature importance is provided to explain the most significant factors that have an impact on the decision to go with a stroller.

In general, reporting and interpreting the results based on the analytical framework developed in Chapter 3 are the themes of this chapter that offers empirical information that guides the design and evaluation of sensory smart strollers among children with ASD.

4.2 Data Exploration

After the data collection by the questionnaire, the data were imported by an Excel file created by using the Google Forms. The second row of the file was the one indicated as the header to make sure that all survey questions were read as column names correctly. The first step was to examine the data set to know its structure, dimensions and quality. The data were 300 responses comprising of 84 variables, including child attributes, sensory experiences, caregivers perception, technology acceptance, and stroller appropriateness.

Dataset shape: (300, 84)

	Unnamed: 0	Child gender	Child's age	How professionals described your child's current level of support?	The age of child showing autistic traits?	How does your child usually communicate with you in daily life?	Your child's walking and mobility characteristics when going out?	How often do your child use Snoezelen room(Calm room)?	Snoezelen room is effective for my child.	The cost of using Snoezelen room is high for me.	...	I am willing to use a sensory smart stroller which can improve my child's comfort.	I am willing to use this type of stroller in high-stress outing situations.	I am willing to pay extra for a stroller with sensory functions.	I am willin I recommen a senso sma stroller others nee
0	1	Male	3~5	L1	< 2	Limited speech	Walks independently	1-2 / Month	Yes	Yes	...	4	4	5	
1	2	Female	6~8	L2	2~3	normal	Walks independently	1-2 / Month	Not sure	Yes	...	5	4	5	
2	3	Female	3~5	L2	< 2	Limited speech	Needs assistance	Never	No	Yes	...	5	4	5	
3	4	Male	3~5	L2	< 2	Limited speech	Walks independently	1-2 / Month	Not sure	Yes	...	5	4	4	
4	5	Female	6~8	L1	2~3	normal	Walks independently	1-2 / Month	Yes	Not sure	...	5	4	5	

Figure 1 Overview of data

A system generated index was eliminated in the first column to eliminate redundancy. The rest of the variables were left to be processed further. In order to increase readability and analytical consistency, original column names in forms of questionnaire were renamed in a systematic way with short, machine-readable labels.

The renaming process made clarity in later steps of preprocessing and modelling but the semantic meaning of individual survey item was maintained.

	Child gender	Child's age	How professionals described your child's current level of support?	The age of child showing autistic traits?	How does your child usually communicate with you in daily life?	Your child's walking and mobility characteristics when going out?	How often do your child use Snoezelen room(Calm room)?	Snoezelen room is effective for my child.	The cost of using Snoezelen room is high for me.	Is it easy to find Snoezelen room near your residence.	...	I am willing to use a sensory smart stroller which can improve my child's comfort.	I am willing to use this type of stroller in high-stress outing situations.	I am willing to pay extra for a stroller with sensory functions.
0	Male	3~5	L1	< 2	Limited speech	Walks independently	1-2 / Month	Yes	Yes	No	...	4	4	5
1	Female	6~8	L2	2~3	normal	Walks independently	1-2 / Month	Not sure	Yes	Not sure	...	5	4	5
2	Female	3~5	L2	< 2	Limited speech	Needs assistance	Never	No	Yes	No	...	5	4	5
3	Male	3~5	L2	< 2	Limited speech	Walks independently	1-2 / Month	Not sure	Yes	No	...	5	4	4
4	Female	6~8	L1	2~3	normal	Walks independently	1-2 / Month	Yes	Not sure	Not sure	...	5	4	5

Figure 2 1st column removal

```
Columns renamed successfully!
```

```
df_clean.head()
```

	child_gender	child_age	support_level	autism_trait_age	communication	mobility	snoezelen_freq	snoezelen_effective	snoezelen_cost	snoezelen_access	..
0	Male	3~5	L1	< 2	Limited speech	Walks independently	1-2 / Month	Yes	Yes	No	.
1	Female	6~8	L2	2~3	normal	Walks independently	1-2 / Month	Not sure	Yes	Not sure	.
2	Female	3~5	L2	< 2	Limited speech	Needs assistance	Never	No	Yes	No	.
3	Male	3~5	L2	< 2	Limited speech	Walks independently	1-2 / Month	Not sure	Yes	No	.
4	Female	6~8	L1	2~3	normal	Walks independently	1-2 / Month	Yes	Not sure	Not sure	.

Figure 3 Columns renaming

A data completeness analysis was conducted using a missing value check. The open-ended question (openfeatures) had the highest number of missing responses, 285 to be precise which was anticipated considering the nature of this question was optional. There was minimal or no missing data in the other variables. Duplicates checking ensured that the dataset did not contain any repeated responses. In general, the quality of the data was found to be satisfactory to proceed with the analysis.

```
 MISSING VALUES PER COLUMN:
open_features       285
child_age             0
child_gender          0
autism_trait_age      0
communication         0
                    ...
cg_age                0
cg_education          0
income                0
area                  0
relation              0
Length: 83, dtype: int64

 NUMBER OF DUPLICATED ROWS:
0
```

Figure 4 Missing values overview

Lastly, the individual value of the variables were also analyzed to point inconsistencies in the formatting and response classification. Small problems like the inclusion of additional spaces, discrepancies in symbols, and type differences were eliminated through standardizing the text entries. Responses in time and price were reformatted in order to be consistent. Also, suitability of strollers responses were grouped together, and those answered as Not sure were eliminated to enable binary classification. There were also typographical discrepancies in the usage frequency and household income categories which were fixed. These actions guaranteed that the data has been clean, consistent and it will be further encoded and analysed using machine learning.

4.2.1 Exploratory Data Analysis (EDA)

Child gender distribution in this sample indicates a surplus of males (170) over females (77), which is the same pattern as previously reported that autism spectrum disorder is more often diagnosed among boys (Centers for Disease Control and Prevention, 2021; World Health Organization, n.d.). This difference can indicate that,

in this sample, boys are more prone to visible characteristics under consideration, whereas girls may be having covert symptoms that are not often detected.

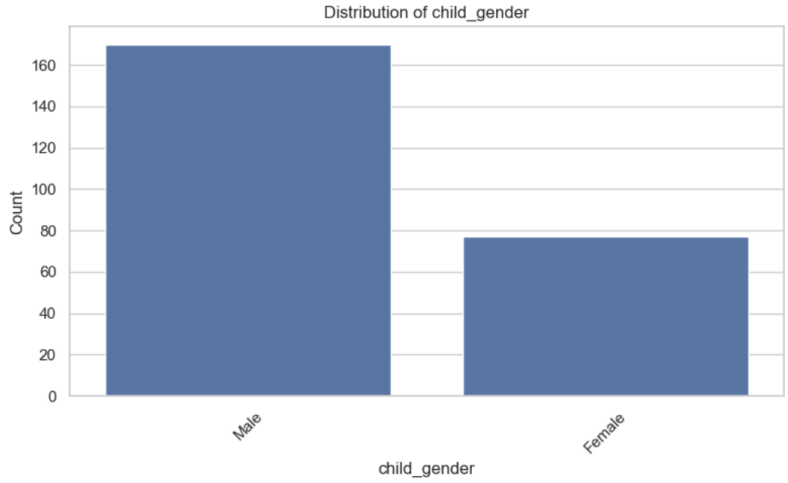


Figure 5 Gender distribution

Child age analysis reveals that the majority of children were in the early childhood span (age 3-8) (179 children total) which is a crucial stage in the developmental intervention. There are less children below 3 years (29) and over 12 years (15) indicating that the data is representative among children who are usually most diagnosed and have early intervention at those ages (Marco et al., 2011).

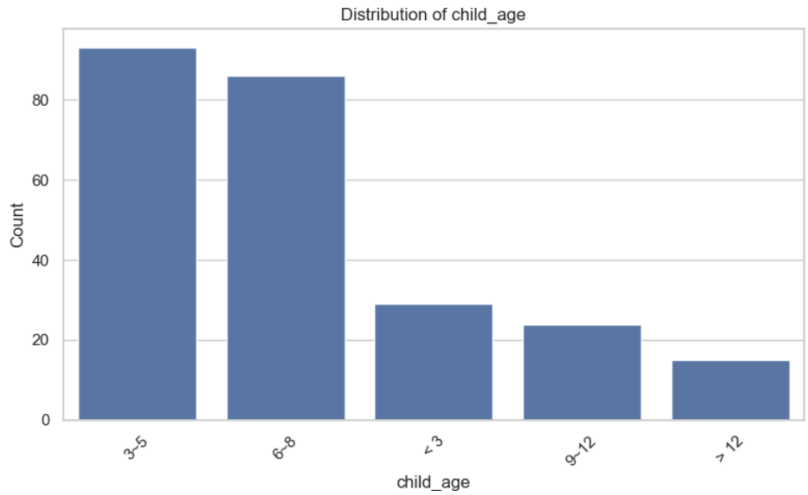


Figure 6 Age distribution

The support level data reveal that the high level of support (L1: 116) and moderate level (L2: 113) are needed by the significant number of children, with only some children at the L3 (15) or unsure level (3). The implication of this distribution is

that the majority of children in the sample encounter significant challenges on a daily basis and could use specific help through the social and outdoor activities (Johansen et al., 2016).

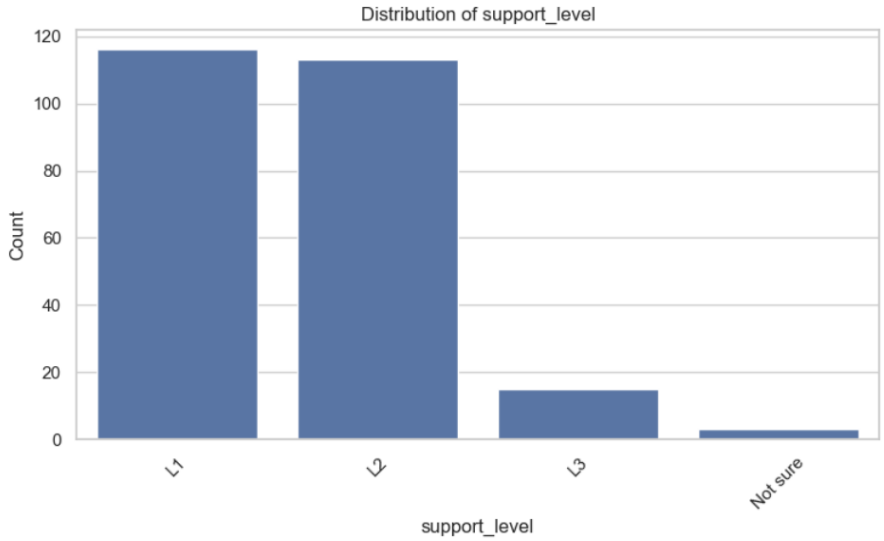


Figure 7 Support level overview

Concerning the age of the initial notice of autistic characteristics, the majority of children displayed indications prior to age 3 (134 below 2 years, 88 between 2 and 3 years). Early emergence of traits complies with the focus on the early detection that is important in the context of effective interventions and assisting caregivers with managing behaviors (Gonderson et al., 2024).

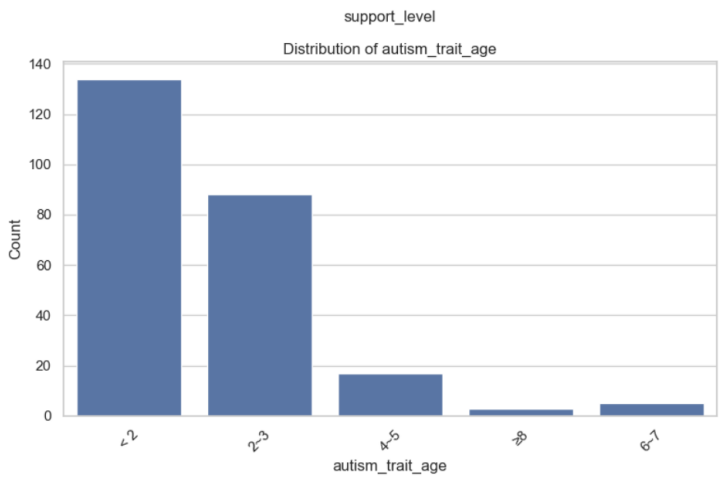


Figure 8 Autism trait age distribution

To communicate, the data shows that there are a high amount of children with limited speech (120), almost equal amount of children that can speak normally (117), and a small amount that is non-verbal (10). This difference implies that caregivers face different communication requirements, and this affects the approaches towards interaction, sensory support and outdoor activities (Johansen et al., 2016).

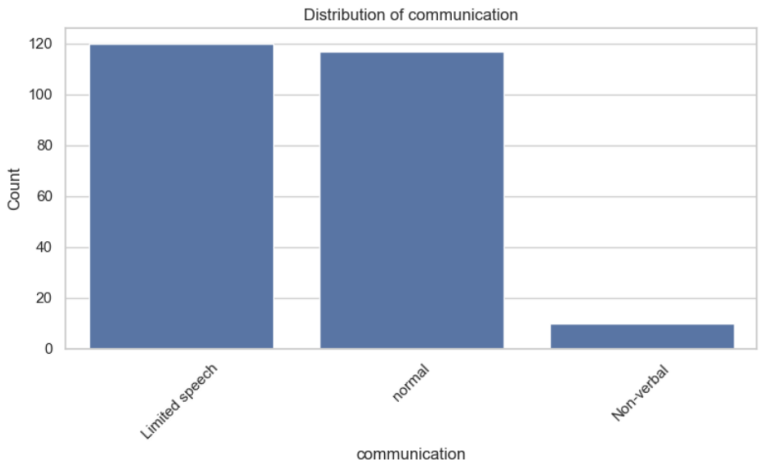


Figure 9 Communication distribution

Lastly the mobility features show that most of the children (193) walk freely with others (50) having to be assisted and very few having physical constraints (4). Independent mobility will probably impact the amount of supervision required, as well as the kinds of outdoor experience the caregivers can safely offer. Outings can cause increased stress among the children who need some help, which is why such supportive tools as the strollers or assistive devices are necessary (Martins et al., 2012).

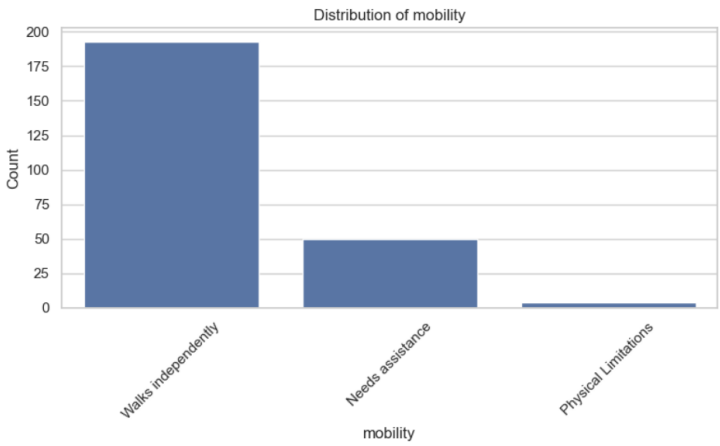


Figure 10 Mobility distribution

All in all, the sample represents a representative picture of children with autism, including early age onset, female-to-male ratios, diverse communication skills, and support requirements, which are all included to design interventions, sensory equipment, and support models to be applied by caregivers.

4.3 Correlation between variables

Correlation analysis was a study that provides the relationship between the target variable stroller suitability and other variables in the dataset. Sensoryoutdoor (0.185), confidencehandling (0.155), snoezelen frequency (0.153), and outdoorduration (0.151) had the strongest positive correlated results. Such findings indicate that children who have a higher rate of sensory discomfort out of the house, whose parents are more confident in managing the difficulties, and who visit Snoezelen rooms more often or spend a longer time outdoors are more likely to have a stroller that is more appropriate to their requirements (Glod et al., 2015).

On the other hand, it was observed that the negative correlations were found with stressstressed (-0.236), autismtraitage (-0.202), stresslimit (-0.140), stresseffort (-0.120), and sensorlight (-0.118). It means that greater stress levels of caregivers, later age of initial autistic characteristics, and light sensitivity are linked with reduced levels of stroller suitability rating (Deng & Rattadilok, 2022). This result supports existing evidence discuss in Chapter 2 that caregiver stressed significantly affect the stroller suitability.

These trends can also be identified through the heatmap visualization, with the variables connected to caregiver confidence, outdoor exposure, and sensory challenges indicating a positive correlation with the stroller suitability and the variables connected with stress-related factors and some sensory sensitivities displaying a negative correlation. These lessons give the idea on what factors are best applied in developing a stroller that best fits the needs of both child and caregiver.

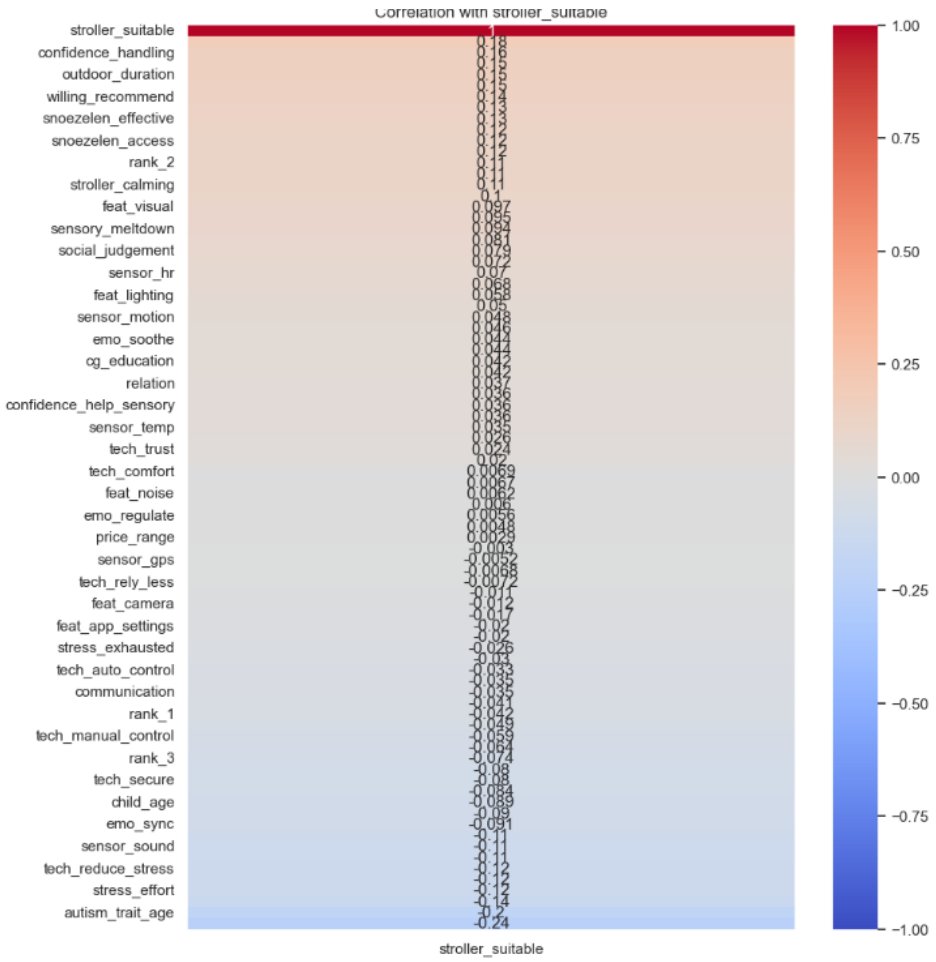


Figure 11 Heatmap

4.4 Distribution of Target Variable

The target variable, strollersuitable, was studied to determine the distribution of the classes in the data set. The findings indicate that higher percentage of the caregivers indicated that a proper stroller is there (167 responses) than those who indicated otherwise (80 responses). It means that the balance of classes is moderate, and positive answers are more widespread.

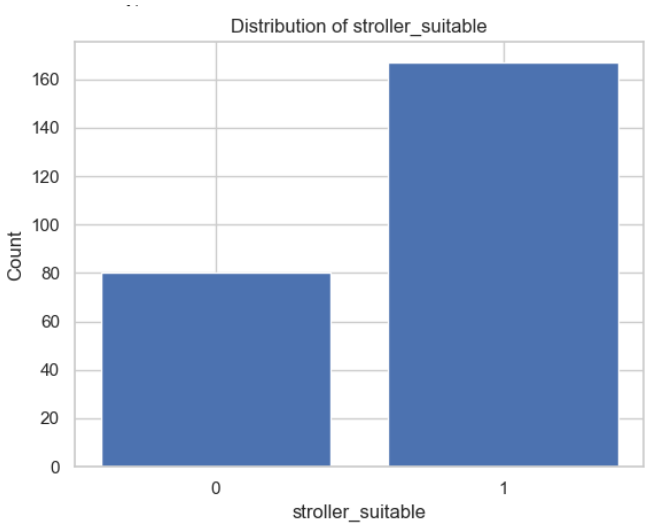


Figure 12 Target class distribution

The dataset was divided into training and testing sets in the ratio of 80:20 after defining the target variable of the strollersuitable. The stratified sampling was utilized to maintain the representativeness of the target variable of both the subsets in terms of class distribution. The initial data as depicted by the proportions had some positive answers of about 68 percent and the negative answers of about 32 percent. This was a steady distribution in both the training and test sets, which means that the split was not a source of sampling bias. To maintain this balance, it is necessary to make sure that the results of model evaluation can be considered as the true reflection of the real-world trends of data and are not influenced by the unbalanced representation of the classes.

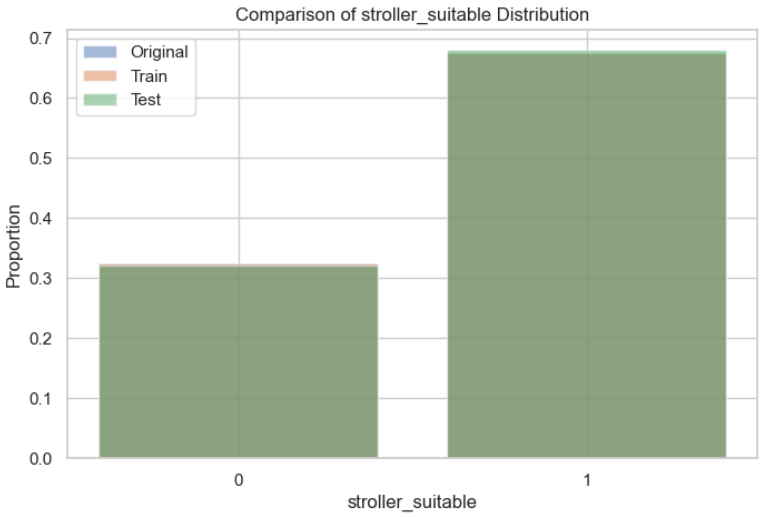


Figure 13 Data split

4.5 Model Training and Performance After Class Balancing Using SMOTE

To overcome the problem of the imbalance of classes in the target variable (strollersuitable), SMOTE was utilized on the training dataset, which led to the creation of an equal measure of classes. Three classification models were then trained on the resampled data resulting in Logistic Regression, Random Forest and Gradient Boosting all tested on the original test set.

```
stroller_suitable
0    133
1    133
```

Figure 14 Data distribution post SMOTE

In general, all three methods (Logistic Regression, Gradient Boosting and Random Forest) had the highest accuracy of 78 and 74 respectively. This implies that the simpler and moderately complex models had similar levels of predicting the suitability of strollers in this dataset.

On the models with respect to the classes, all the models were more accurate on predicting positive class (suitable stroller) which results in line with the higher prevalence of cases in the data set. Logistic Regression and Gradient Boosting showed a trade-off at par between accuracy and recall of both classes indicating more consistent and reliable predictions. Their confusion tables have less instances of misclassification than the Random Forest, especially the negative class.

Conversely, the Recall of the positive type was high in the case of the Random Forest, which was not able to identify the negative cases appropriately, a fact that was indicated by greater number of false positives. This implies that it was clear that Random Forest was not effective in classifying inappropriate cases but it was effective in classifying appropriate strollers.

The comparable output of Logistic Regression and Gradient Boosting implies that the correlations between the input variables and suitability of the strollers can be moderately nonlinear yet not very complicated (Madububambachu et al., 2024). Also, the lack of high accuracy in all of these models suggests that stroller suitability is a

combination of several interacting variables, not all of which may be adequately described using linear correlations.

In sum, the most reliable baseline models were determined to be Logistic Regression and Gradient Boosting as they provide a reasonable combination of intelligibility and predictive accuracy. The results obtained support the idea of the further refinement of the models and the levels of feature interpretation in the later analysis.

```
--- Logistic Regression ---
Accuracy: 0.78
Classification Report:
              precision    recall  f1-score   support

           0       0.67      0.62      0.65        16
           1       0.83      0.85      0.84        34

    accuracy                           0.78        50
   macro avg       0.75      0.74      0.74        50
weighted avg       0.78      0.78      0.78        50

Confusion Matrix:
[[10  6]
 [ 5 29]]


--- Random Forest ---
Accuracy: 0.74
Classification Report:
              precision    recall  f1-score   support

           0       0.67      0.38      0.48        16
           1       0.76      0.91      0.83        34

    accuracy                           0.74        50
   macro avg       0.71      0.64      0.65        50
weighted avg       0.73      0.74      0.72        50

Confusion Matrix:
[[ 6 10]
 [ 3 31]]


--- Gradient Boosting ---
Accuracy: 0.78
Classification Report:
              precision    recall  f1-score   support

           0       0.67      0.62      0.65        16
           1       0.83      0.85      0.84        34

    accuracy                           0.78        50
   macro avg       0.75      0.74      0.74        50
weighted avg       0.78      0.78      0.78        50

Confusion Matrix:
[[10  6]
 [ 5 29]]
```

Figure 15 Performance measure

4.5.1 Confusion Matrix

Following the use of SMOTE, the training data was made balanced, where all of the models learned using the same number of appropriate and inappropriate cases.

Logistic Regression performed moderately and the confusion matrix of the result indicated a fair amount of balance between true positives and true negatives. Nevertheless, it also gave a significant amount of false negatives, which means that it is hard to recognize all appropriate stroller cases. This weakness could be because of the linearity of the model, which limits the capability of forming complex relationships of sensory, emotional, and environmental factors.

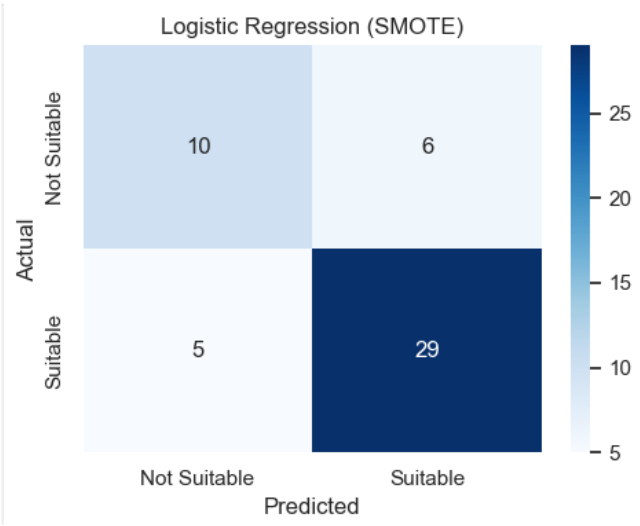


Figure 16 LR confusion matrix

Random Forest had good recall of appropriate strollers, and correctly classified most of the positive ones. But the confusion matrix showed that it had many false positives, like strollers who were not appropriate were mistaken as appropriate. This

bias leads to the fact that this model was inclined to positive forecasts, which might diminish the reliability of the conditions of the inappropriate stroller.

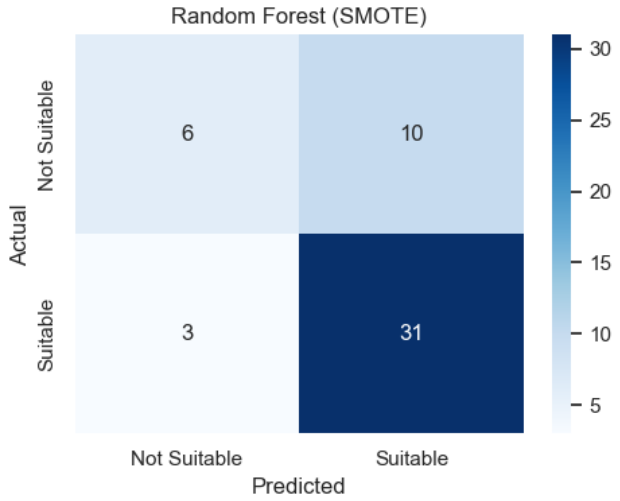


Figure 17 RF confusion matrix

The gradient boosting showed the most equal confusion matrix as compared to the SMOTE based models. It had a similar degree of accuracy to that of Logistic Regression with a lower number of misclassifications in general. False positives and false negatives were relatively low as well, which suggests the more consistent performance in both classes.

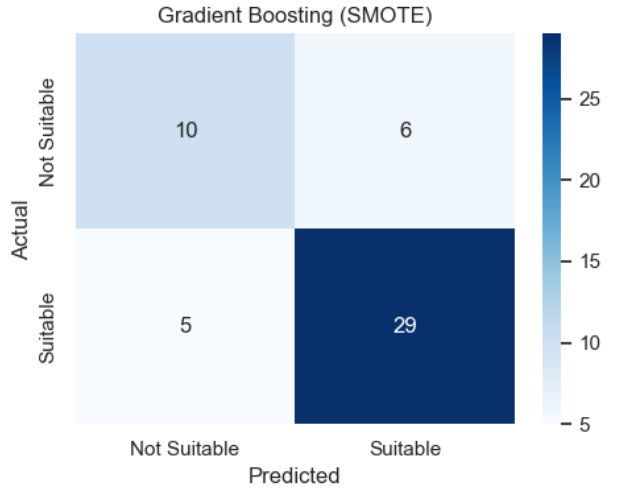


Figure 18 GB confusion matrix

4.5.1.1 Model Training and Performance After Class Balancing Using SMOTE (Hyperparameter Tuning)

In order to further enhance predictive performance, we used grid search Cross-validation, which was hyperparameterized with GridSearchCV with 5-fold stratified cross-validation. The aim of this process was to settle on the best model configurations without compromising the performance in both classes.

Gradient Boosting, as a result of tuning, displayed the best overall performance with an accuracy of 86 which was significantly greater than the Logistic Regression (62) and the Random Forest (72). The Gradient Boosting model also demonstrated the highest equal precision and recalls in both classes, which means that it has a high capacity to identify both appropriate and inappropriate cases of strollers.

The confusion matrix of Gradient Boosting shows that there are fewer misclassifications than the other models especially the negative class. This implies that Gradient Boosting was more efficient in the detection of complex interactions on sensory, environmental, and caregiver-related features. The increased performance suggests that the suitability of strollers is dependent on nonlinear relationships that can be very hard to capture using simple models.

Conversely, the performance of the Logistic Regression deteriorated following tuning especially in the determination of appropriate strollers. This result implies that a purely linear decision line might be inadequate to reflect on the complexity of sensory sensitivity, caregiver stress, and environmental influence of the variable that are represented on the questionnaire data.

Random Forest was found to have moderate results, having a good recall of the appropriate strollers but poor results on the inappropriate cases. Although the ensemble, its low level of detail and divide limitation might have limited its capability in the fine interaction of features than in Gradient Boosting.

All in all, these findings suggest that Gradient Boosting is the best model to use in this research, which has the best predictive accuracy and balanced classification. This helps to conclude that the suitability of a stroller to children with autism is

determined by a complex and interacting factor and not linear relationships. Gradient Boosting, in turn, was chosen as the last model to be further analyzed and interpreted.

```
Training LogisticRegression...
Best params: {'C': 0.05, 'solver': 'lbfgs'}

Training RandomForest...
Best params: {'max_depth': 5, 'min_samples_split': 5, 'n_estimators': 100}

Training GradientBoosting...
Best params: {'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 100}


LogisticRegression Classification Report:
              precision    recall  f1-score   support

           0       0.43      0.62      0.51        16
           1       0.78      0.62      0.69        34

    accuracy                           0.62        50
   macro avg       0.61      0.62      0.60        50
weighted avg       0.67      0.62      0.63        50

Confusion Matrix:
[[10  6]
 [13 21]]

RandomForest Classification Report:
              precision    recall  f1-score   support

           0       0.62      0.31      0.42        16
           1       0.74      0.91      0.82        34

    accuracy                           0.72        50
   macro avg       0.68      0.61      0.62        50
weighted avg       0.70      0.72      0.69        50

Confusion Matrix:
[[ 5 11]
 [ 3 31]]

GradientBoosting Classification Report:
              precision    recall  f1-score   support

           0       0.76      0.81      0.79        16
           1       0.91      0.88      0.90        34

    accuracy                           0.86        50
   macro avg       0.84      0.85      0.84        50
weighted avg       0.86      0.86      0.86        50

Confusion Matrix:
[[13  3]
 [ 4 30]]
```

Figure 19 Performance measure (Fine tuned)

4.5.2 Confusion Matrix

In the latter method, SMOTE was further optimised with hyperparameter tuning to improve the model performance. Tuned Logistic Regression was better than the untuned type especially in the discovery of appropriate strollers. Nonetheless, the confusion matrix still revealed that certain inappropriate cases were misclassified, indicating that there was not much freedom in working with nonlinear patterns.

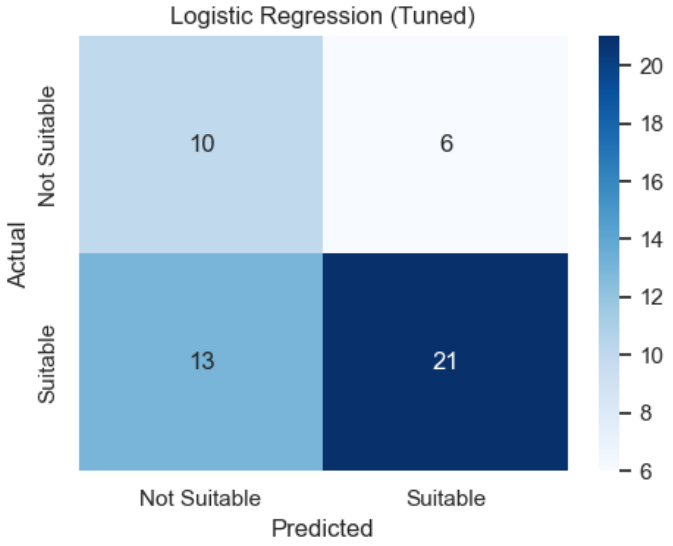


Figure 20 Tuned LR confusion matrix

Controlled tree depth and split criteria in Tuned Random Forest led to less overfitting. Although the recall of appropriate strollers was high, weaknesses were still

observed in the confusion matrix that would denote false positives as the appropriate cases were incorrectly identified.

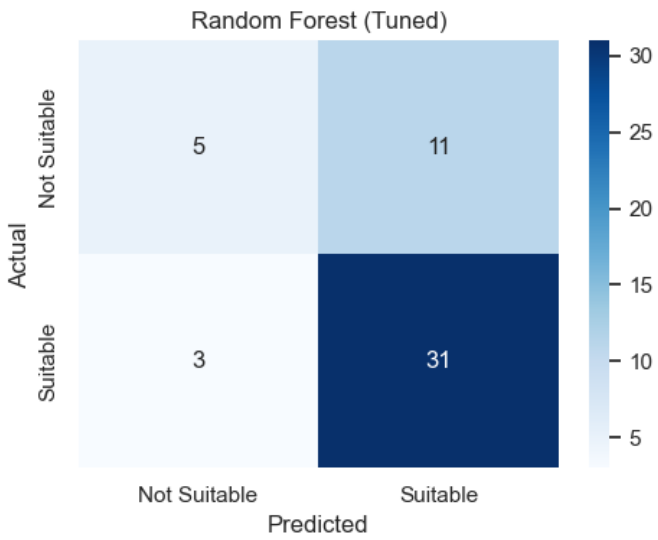


Figure 21 Tuned RF confusion matrix

Tuned Gradient Boosting had the best overall performance. It indicated the greatest number of correct predictions on each of the classes and lower false positives and false negatives in its confusion matrix. This implies that, the tuned model had a greater capacity to detect the subtle interactions between caregiver stress, sensory sensitivity and environmental conditions.

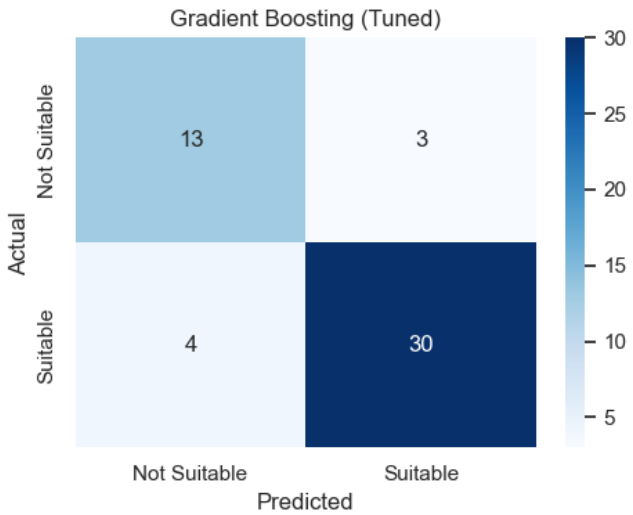


Figure 22 Tuned GB confusion matrix

4.6 Gradient Boosting Feature Importance

In the comparison of both methods, Gradient Boosting was always the most robust model, and the optimized version produced the best overall results. The confusion matrix analysis demonstrated that, once the issue of class imbalance was reduced with the help of SMOTE, the hyperparameter tuning enhanced one more in order to distinguish the appropriate and inappropriate cases of strollers with the help of narrowing down the decision boundaries and decreasing the cases of misclassification.

These results indicate that there are complex interactions between several factors whose suitability in children with autism is not linear. The combination of ensemble learning techniques and proper data balancing and model optimisation makes them more appropriate in order to reflect the complexity of the variables used (caregiver, sensory and environmental).

The twenty features that were ranked as the most important ones by the Gradient Boosting model in terms of influencing the suitability of the stroller to children with autism. The findings indicate that the sensory attributes of the children and the factors associated with the caregiver are very critical factors that help in deciding whether a stroller is the right option on the outdoor and daily use situations.

Sensoryoutdoor is the most significant one, which means that the sensory reaction of a child in the outdoor activities is the most effective predictor of the stroller suitability. This demonstrates that stroller design should be more concerned with trying to deal with sensory difficulties that are encountered in the outdoor setting including noise exposure, light exposure, temperature exposure, and mobility (Marco et al., 2011). Next in line is autismtraitage, an opinion that indicates that age at which autism traits were determined affects the needs of the strollers, and may be a sign of the developmental and sensory regulation capabilities at different developmental stages (Gonderson et al., 2024).

Some of the caregiver stress variables such as stresseffort, stressstressed, stressexhausted, stressanxious, and stressavoid are among the most significant features.

This implies that the suitability of a stroller is not only dependent on the sensory profile of a child, but also the emotional and physical burden of the caregivers. The increased stress levels are probably related to the increased necessity of supportive features in strollers that will minimize the efforts, make them easier to control, and make the outdoor activities easier to manage (Fang & Ping, 2022).

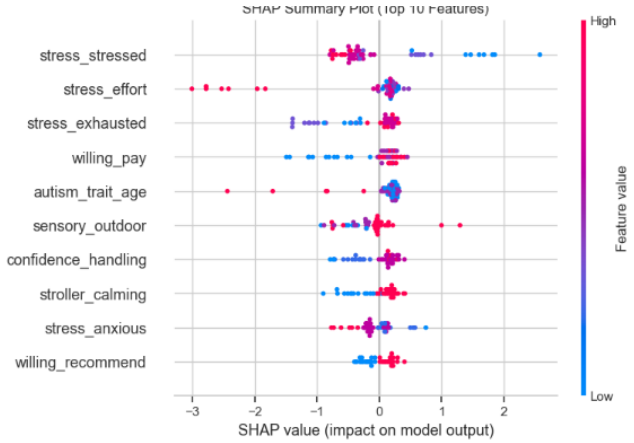
Sensory specific variables like sensorylight are another pinpoint in the significance of regulating environmental stimuli. Sensitivity to light implies that the stroller appropriateness in children with autism could be greatly enhanced by means of adjustable lighting, shading or visual filters. Snoezelen-related features such as snoezeleneffective, snoezelencost and snoezelenaccess suggest that the experiences of the caregivers with the sensory calming environments affect the expectation about steller performance.

These results indicate that caregivers appreciate sensory calming qualities and evaluate the stroller appropriateness in noticeable effectiveness, availability, and sturdiness of the sensory interventions (Patil & Kaple, 2023). The variables that are related to technology like sensortemp, sensormotion, sensorair and techrelyless indicate the importance of sensor based monitoring and adaptive features. The significance of these variables implies that the caregivers think that value in the strollers would be in the capability of the strollers to perceive the surrounding environments and react to them, as well as strike a balance between dependence on technology (Almutairi et al., 2022)

Willingpay, confidencehandling, and emosync reflect on the user acceptance and their behavioral intention. These features indicate that caregiver confidence in their ability to use the stroller, emotional control with the child, and the desire to invest in assistive technology also influence the suitability of the strollers. This further supports the fact that suitability is not technical alone but psychological and economic. Lastly, the variables of practical usage like strollerfreq and background factors like ogeducation give an indication that practical usage pattern and caregiver knowledge is a contributive factor to suitability perceptions as well.

All in all, the findings of the Gradient Boosting feature importance indicate that the suitability of strollers to children with autism is a multidimensional construct

that involves child sensory reactions, caregiver stress and confidence, sensory management experience, and technology acceptance. These results confirm that machine learning models based on questionnaires could inform the design of sensory responsive strollers that consider the comfort and care of children along with the interests of caregivers.



Top 10 Features by Mean Absolute SHAP Value:

	Feature	Mean |SHAP Value|
22	stress_stressed	0.659756
21	stress_effort	0.505887
23	stress_exhausted	0.403878
73	willing_pay	0.344527
3	autism_trait_age	0.307716
18	sensory_outdoor	0.278402
31	confidence_handling	0.262175
13	stroller_calming	0.238756
24	stress_anxious	0.223869
74	willing_recommend	0.192266

Figure 23 SHAP analysis

4.7 Conclusion

Chapter 4 showed the data exploration, correlation, and machine learning results to determine factors that influence the suitability of the stroller to children with Autism Spectrum Disorder. The results revealed that child sensory reactions especially outdoors, and caregiver-related variables including stress, confidence and experience with sensory interventions had a significant impact on the use of the strollers. The Gradient Boosting model was found to be the most useful predictive model, as it detects the multisensory interaction among sensory sensitivities, the environment, and the caregiver perceptions (Alateyat et al., 2022). The feature importance analysis revealed that the suitability of the stroller is multidimensional, and it is very important to consider the child comfort, sensory management, caregiver support, and technology

acceptance. These results offer a good empirical base to develop sensory receptive strollers that are specific to the special needs of children with ASD and their caregivers.

CHAPTER 5

CONCLUSION AND FUTURE WORKS

5.1 Introduction

The chapter introduces and describes the findings of the machine learning analysis performed in the framework of this research, which is expected to investigate the variables that influence the suitability of a stroller to children with Autism Spectrum Disorder using questionnaire-based data. This chapter is aimed at interpreting the performance of the chosen machine learning models and conducting a review of the main variables that lead to the results of stroller suitability.

The discussion focuses on the interpretability of models as opposed to predictive accuracy by itself and the confusion matrix and feature importance outcomes are specifically highlighted. In this way, one will have a better idea of how sensory sensitivities and stress levels recorded by caregivers, environmental conditions, and technology-related perceptions affect the suitability of the strollers. Out of the analyzed models, the Gradient Boosting model was chosen to be analyzed more thoroughly because of its good results and the possibility to predict complicated relationships in the data (Alateyat et al., 2022).

The most important factors influencing the predictions of stroller suitability are determined with the help of feature importance analysis. The findings are presented in light of the available literature on the topics of sensory processing, caregiver stress, and assistive mobility of children with autism. The results of this chapter have some practical implications on the design of sensory responsive smart strollers and form a basis of the conclusions and recommendations made later in the chapter.

5.2 Discussion of Key Findings

The key findings of Chapter 4 are discussed in this section by interpreting the data of exploratory analysis, correlation analysis, machine learning performance, and feature importance in terms of suitability of strollers to children with Autism Spectrum Disorder. The analysis of the machine learning through the questionnaire is discussed in terms of the effect of child sensory properties, caregiver-related perceptions as well as technology-related perceptions on the suitability of strollers.

The exploratory data analysis revealed that it is mainly a group of young children aged 3 to 8 with more of the male children in the dataset and moderate to high needs in support. These attributes align with the current trends in autism prevalence and imply that the suitability of using a stroller is the most topical in the first years of childhood when the mobility and the involvement of the caregiver is the greatest. The differences in communication skills and mobility also suggest that the needs of strollers are very different in children, and such solutions should be adaptive and personalized against conventional designs (Centers for Disease Control and Prevention, 2021; World Health Organization, n.d.).

Correlation analysis indicated that suitability of the strollers has a positive relationship with outdoor sensory experiences, confidence of caregivers in managing challenges, and interactions with sensory environments like Snoezelen room. (Patil & Kaple, 2023) study shows that multisensory calming environments can reduce sensory overload and support ASD child's emotional regularly. Such results indicate that strollers are considered more appropriate when they assist children to overcome sensory requirements in the outdoor processes and when the caregivers feel that they can and are assisted by the use of the strollers. Conversely, the negative relationships with caregiver variables that have a correlation with stress factors and light sensitivity depict that increased emotional demand and uncontrollable sensory stimuli decrease perceived stroller appropriateness (Deng & Rattadilok, 2022). It is in line with the idea that the design of the stroller has to consider the sensory comfort and the workload of the caregiver to be effective.

These insights were also supported by the use of machine learning models. Following the correction of the issue of imbalance of classes with the help of SMOTE, Logistic Regression and gradient boosting proved to be consistent predictors, whereas the use of the Random Forest demonstrated some weaknesses in the recognition of inappropriate stroller cases. The fact that Gradient Boosting works better than hyperparameter tuning shows that the suitability of strollers is not brought about by a linear relationship between factors but rather by complex and interactive factors (Madububambachu et al., 2024). Madububambachu's paper shows the idea of mental health-related outcomes require nonlinear ML approaches. The increased confusion matrix success indicates that the better-tuned Gradient Boosting model was more capable of differentiating appropriate and inappropriate stroller conditions to make it the most suitable model to be used in this study (Alateyat et al., 2022). Alateyat shows the ensemble and boosting-based models outperform linear models when dealing with complex sensory-behavioral ASD data.

The Gradient Boosting model provided a better interpretive value through feature importance analysis. The most important feature, sensoryoutdoor, proves that the sensorial reactions of children in the outdoor conditions are the key to the suitability of strollers (Marco et al., 2011). This observation highlights the fact that environmental exposure is very critical in deciding whether a stroller suits the needs of a child. The significance of autismtrait age also implies that the time of development determines the sensory control and mobility needs.

The variables of caregiver stress came out as one of the major predictors which means that the suitability of strollers is closely related to the emotional and physical stress of the caregivers (Deng & Rattadilok, 2022). The requirement of supportive stroller features becomes more eminent when care givers are more burdened with greater effort, anxiety, exhaustion, or avoidance. This underscores the fact that the suitability of the strollers is not only a child focused idea but also includes the well being and usability of the caregiver (Fang & Ping, 2022).

The applicability of the sensory-conscious and adaptable strollers can be supported by sensory-specific and technology-related characteristics such as sensitivity to light and perceived usefulness of the environmental sensors (Almutairi et al., 2022). The Snoezelen-related variables indicate that caregivers think that the

calming sensory-related strategies are important, and their expectations apply to the design of the strollers (Patil & Kaple, 2023). There is also willingness to pay, confidence with handling, emotional coordination and frequency of stroller usage, which suggests that the perceptions of suitability also depend on acceptance, affordability and practicality in real-world application.

On the whole, the results indicate that the suitability of strollers to children with autism is a multidimensional outcome that depends on sensory experiences, the stress and confidence of caregivers, previous experience with sensory interventions and attitudes to assistive technology. The findings confirm the application of questionnaire-based machine learning as a useful tool in determining the important aspects, that could be used to inform design of sensory-responsive strollers to accommodate children and caregivers (Alateyat et al., 2022).

5.3 Limitations

Although the given study represents an invaluable contribution to the analysis of factors influencing strollers appropriateness among children with Autism Spectrum Disorder through questionnaire analyses supported by machine learning, it has a number of limitations that must be noted.

The dataset is first founded on all caregiver-reported questionnaire responses, which can potentially create subjectivity. (Glod et al., 2015) explicitly discuss variability and subjectivity in sensory processing reports. Personal experiences, expectations or inaccuracy in recall may have an effect on the way caregivers regard their child and their sensory sensitivities, stress or the use of strollers. This reduces the objectivity of input data and could have an impact on the generalizability of the predictive models.

Second, the sample of 300 responses, although adequate to analyze the initial data, might not reflect the whole range of children with ASD of various age categories, cultural diversity or support needs. As a result, the results might not be sufficiently

representative of the population of caregivers and children with autism in general (Glod et al., 2015).

Third, despite the best predictive results, Gradient Boosting is dependent on the variables that are presented in the dataset. The aspects that could not be measured by the questionnaire such as certain environmental conditions, mood fluctuations in the child, or the actual behavior of using strollers were not taken into account. This implies that there were some applicable determinants of stroller suitability which may have been missed.

Fourth, the research aims at predictive modelling and not experimental validation (Fang & Ping, 2022). Although the models determine the major features that are related to the suitability of the strollers, the testing of the stroller designs according to the predictions has not been conducted in reality. Thus, the effectiveness of proposed design changes is not tested in practice.

Lastly, its research mainly focuses on outdoor sensory aspects and the stress on the caregiver whereas other factors, including stroller durability in the long run, expenses, and the availability of assistive technologies in the area were not thoroughly covered. These also might have an effect on the applicability of strollers in reality.

Identification of such limitations puts the findings into perspective and show future research directions in which the future research can be improved.

5.4 Future Works

Based on the results and weaknesses of this paper, it is possible to suggest some recommendations on how future studies on the suitability of strollers among children with Autism Spectrum Disorders (ASD) based on machine learning and caregivers can be conducted. First, the research studies in the future should focus on increasing the sample size and diversity of the participants. Caregivers and children in various geographical regions, socioeconomic backgrounds, and cultural contexts should be

included to enhance the generalization of predictive models and make the subsequent suggestions on the design of the strollers inclusive and universal.

Second, real-time sensor information on strollers used in the predictive framework could offer a more objective and dynamic perspective on sensory issues (Deng & Rattadilok, 2022). As one such example, information about light, sound, movement, and temperature gathered during practice of using strollers may supplement questionnaires filled in by caregivers, which would make the model more accurate and less subjective.

Third, longitudinal research could be conducted in future to investigate how suitability of strollers varies with time as children develop or their sensory sensitivities vary (Gonderson et al., 2024). This would assist in the development of the strollers that may either be adjusted or tailored based on the developmental phases or any other requirements.

Fourth, it is suggested to test stroller prototypes which are predicted by machine learning through experimental validation of the prototype. The model-based recommendations can be tested in real settings using strollers with the anticipated sensory qualities to determine whether they are working to enhance child comfort, confidence in caregivers, and the general usability.

Fifth, cost constraint, accessibility, material durability, and ease of maintenance should be other factors that can be included to further narrow the design recommendations. This is a whole package that will have strollers that are not only sensory reactive but also be practical and cheap to families.

Lastly, involving more sophisticated methods of machine learning, like deep learning or ensemble-based approaches, which consider multiple data modalities, like video observations, wearable sensors, and caregiver input, could be explored in the future (Madububambachu et al., 2024). This would be able to improve the predictability of models and it would be possible to develop actually personalized, adaptable smart strollers in children with ASD.

The research can close the gap between predictive modelling and practical application of children with autism and their caregivers by implementing the following recommendations.

Overall, these results align with the literature on sensory processing and caregiver stress in autism, and assistive technology development for children with disabilities who are highly sensory sensitive, highlighting that mobility solutions need to support both child sensory needs and caregiver usability (Marco et al., 2011; Fang & Ping, 2022).

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